

19-205-0305 Metallurgy and Material Science

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Materials that changed the history:-

- ❖ Wood
- ❖ Ceramics
- ❖ Fiber & Cloth
- ❖ Bronze
- ❖ Iron & Steel
- ❖ Glass
- ❖ Paper
- ❖ Plastic & Rubber
- ❖ Aluminum
- ❖ Semiconductors



Human civilization through materials

- **Stone Age**
- **Bronze Age**
- **Iron Age**
- Glass Age
- Steel Age
- Aluminum Age
- Plastic Age
- Silicon Age
- Designed/Architected Materials Age

Material Science & Engineering

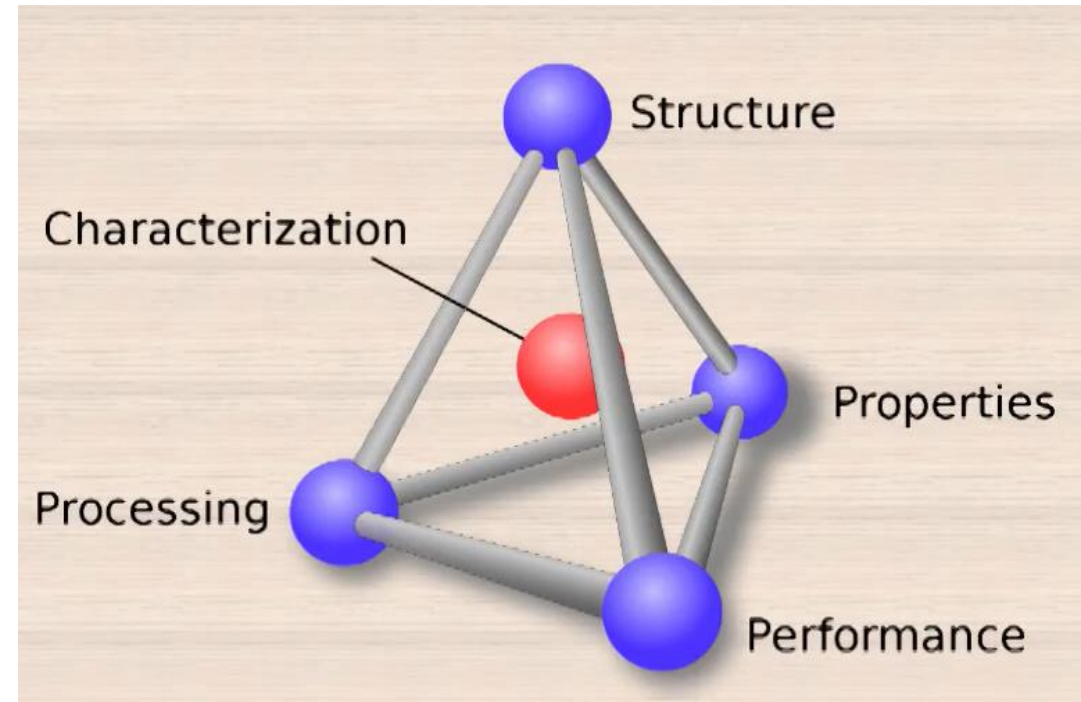
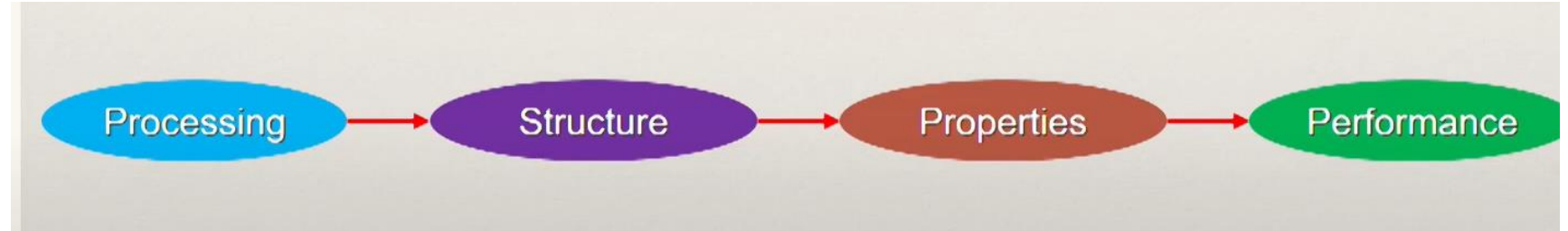
Material Science

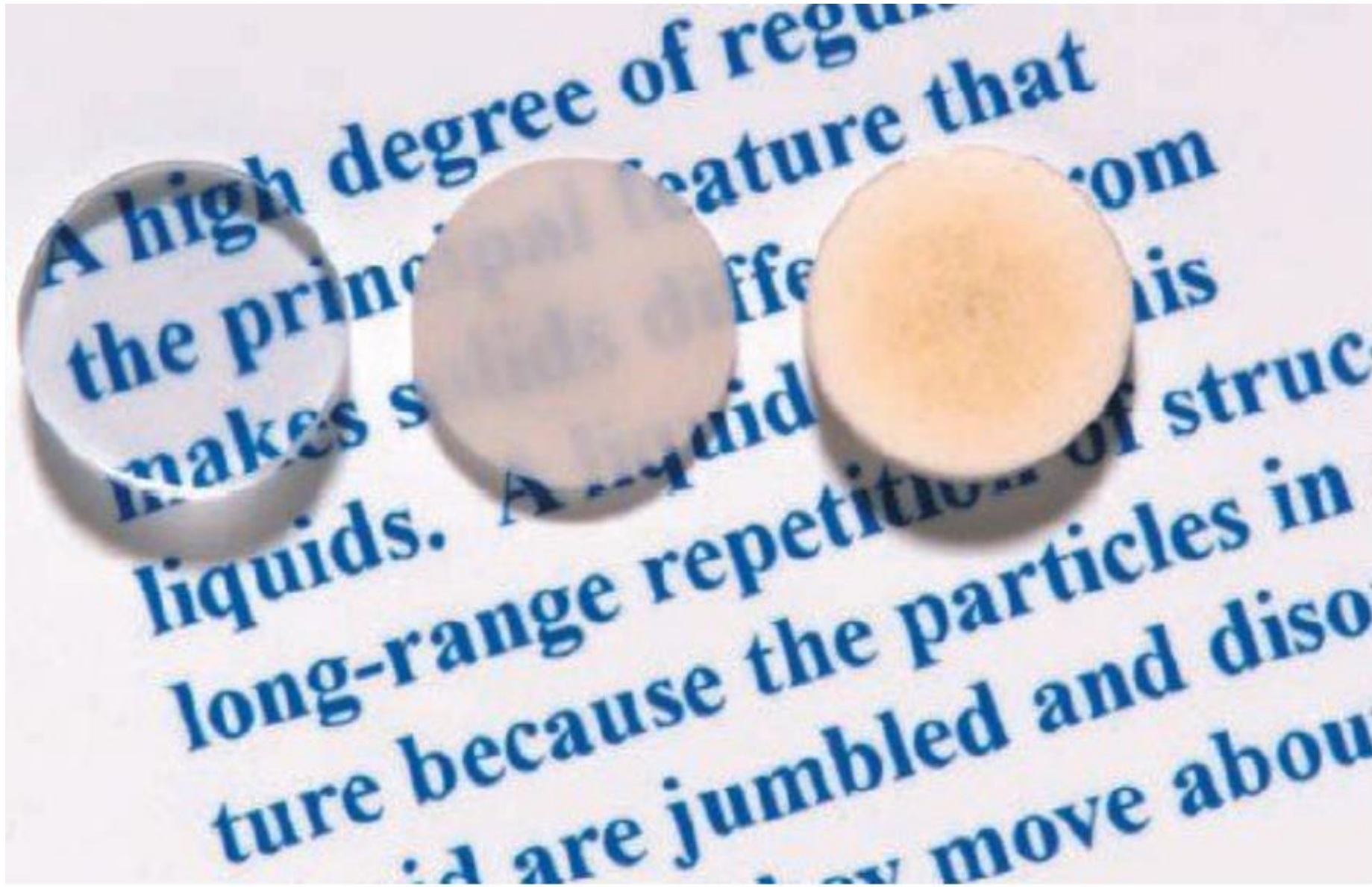
Materials Engineering

Structure

Property

Processing and performance





1 Three thin disk specimens of aluminum oxide that have been placed over a printed page in order to demonstrate their differences in light transmittance characteristics.

Levels of Structure

Depending on the level, we can classify the structure of materials as:

Macrostructure

Microstructure

Substructure

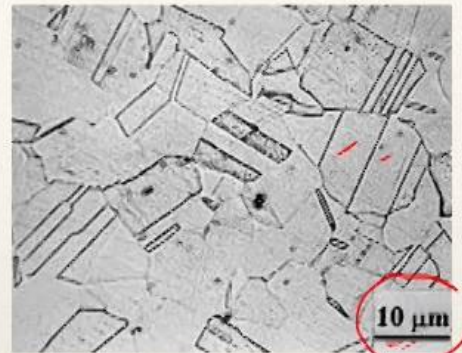
Crystal structure

Electronic structure

Nuclear structure.

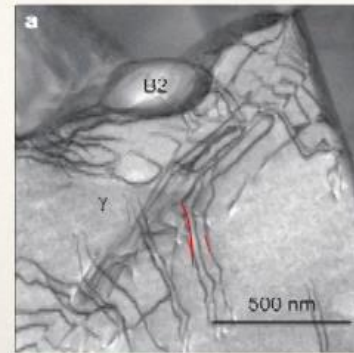


http://www.chaparraltech.net/sintered_metals.html



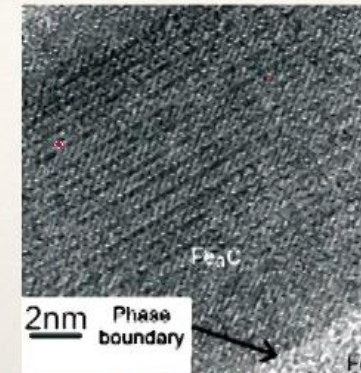
[Wikipedia](#)

Microstructure



http://www.nature.com/nature/journal/v510/n7537/fig_tab/nature14144_13.html

Dislocations



http://www.tl.uni-kiel.de/matwis/amat/iss/kap_5/illustr/5_3_3.html

Individual atoms

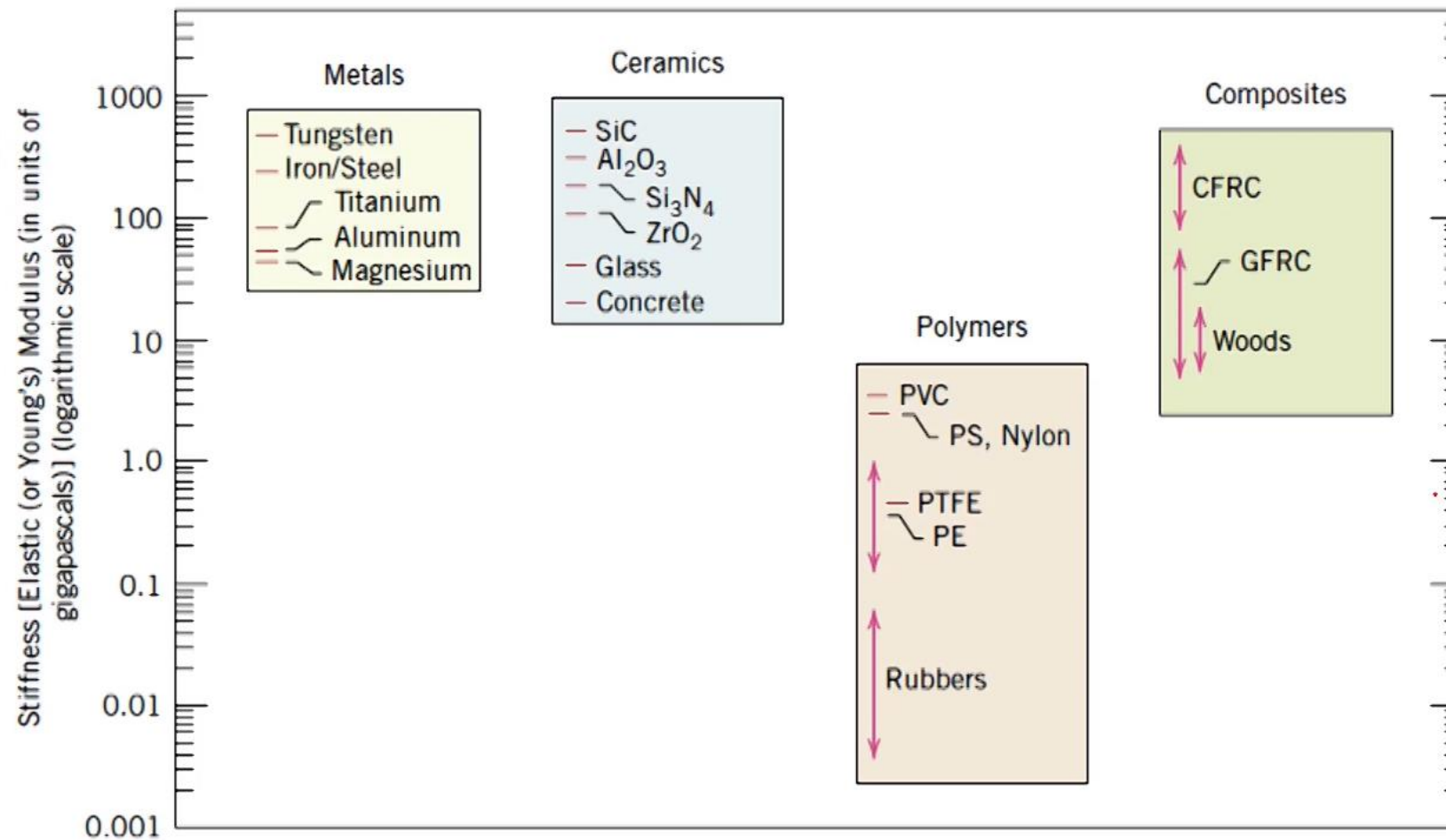
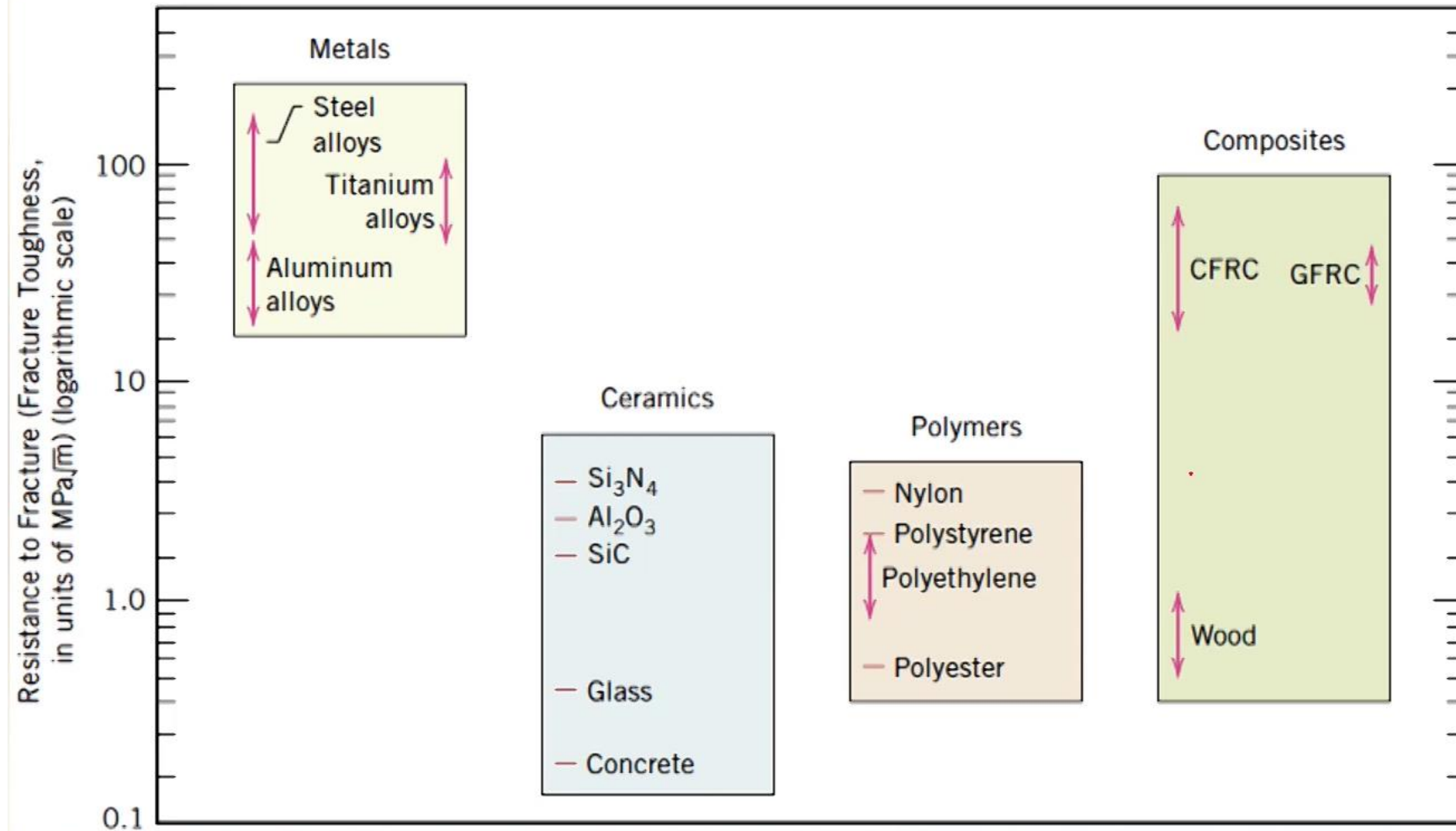
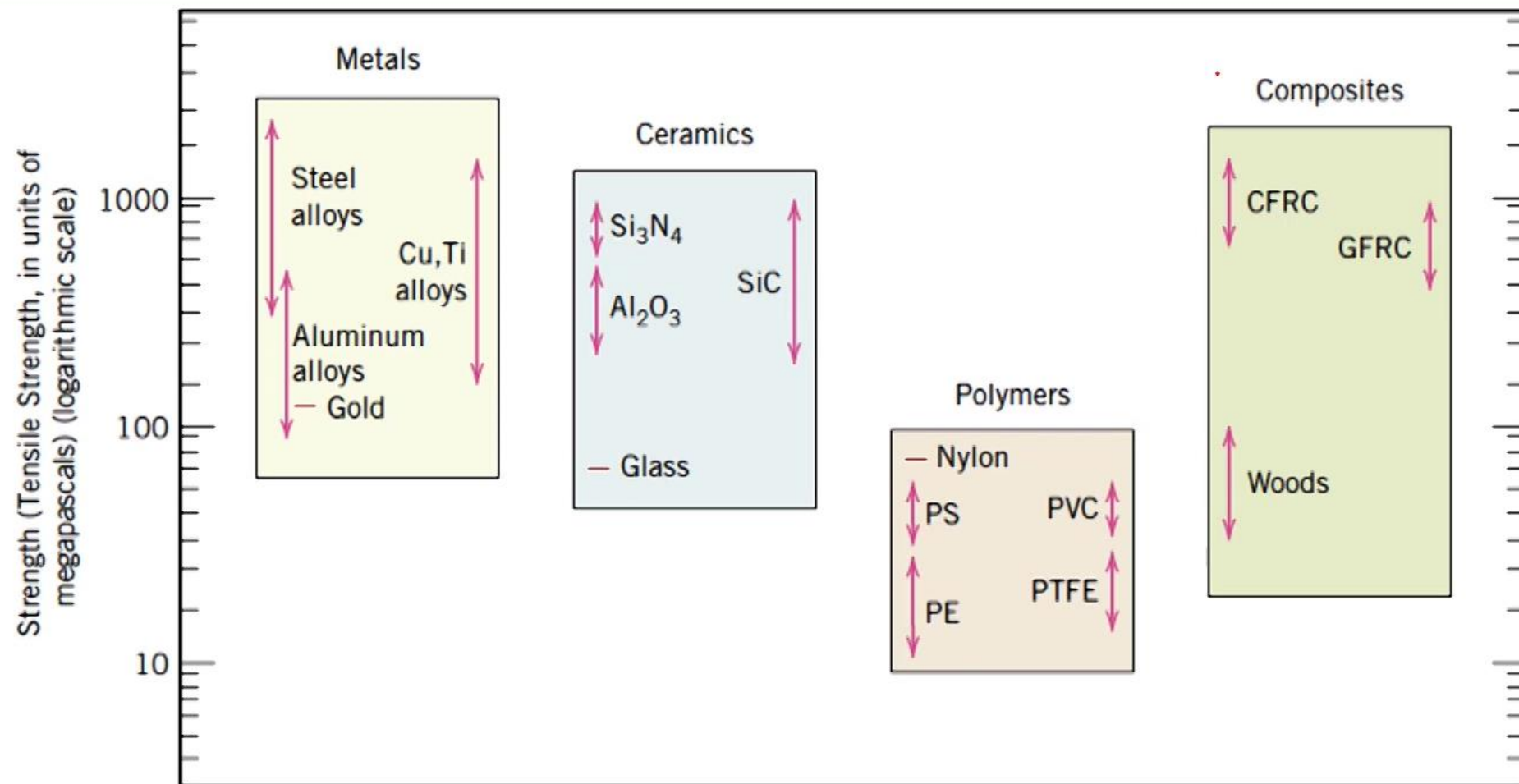
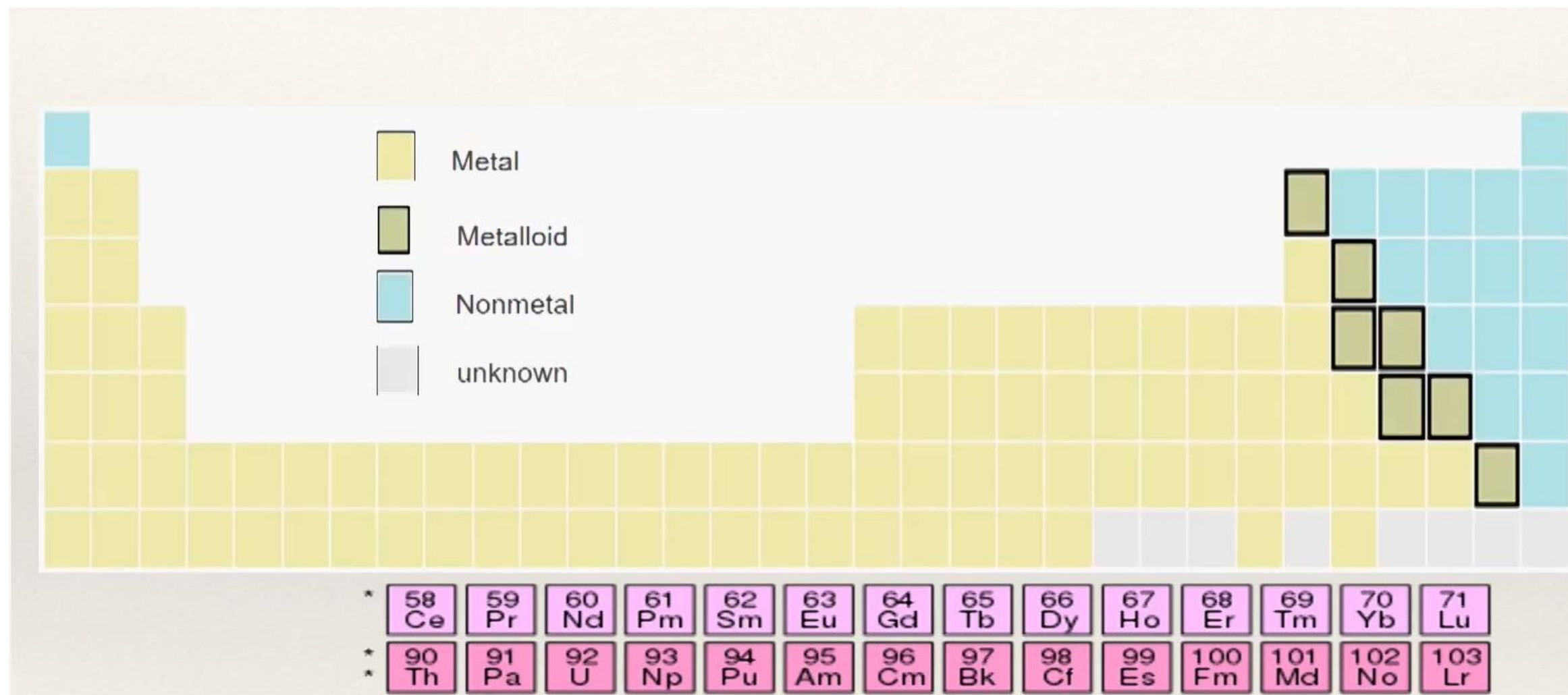


Image source: Callister, Chapter: 1.4, Classification of Materials

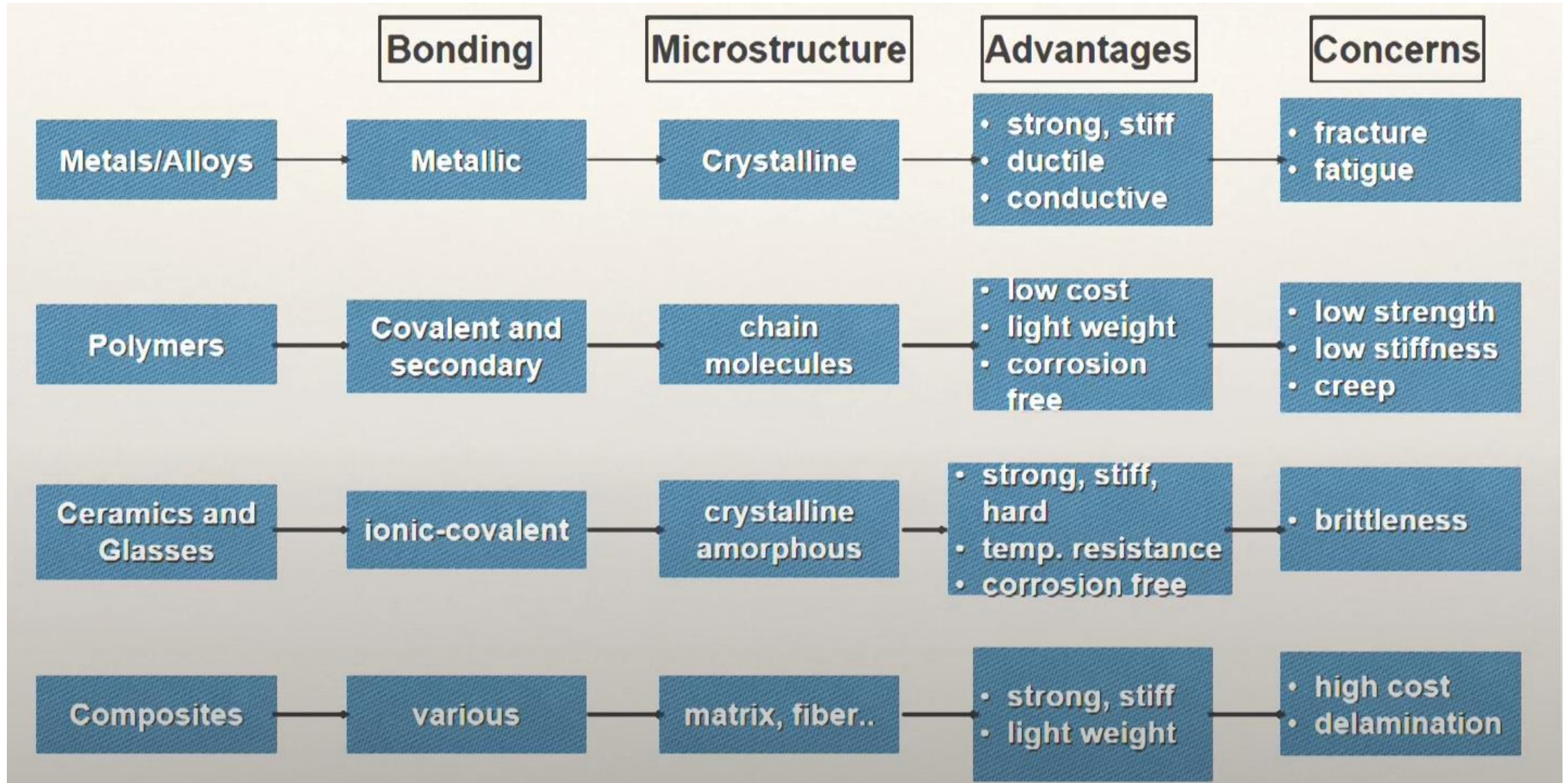




Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 La *	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	89 Ac *	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
				* 58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
				* 90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	



Classification of Materials



Advanced materials

- ❖ Semiconductors
- ❖ Biomaterials
- ❖ Smart materials
- ❖ Nanomaterials

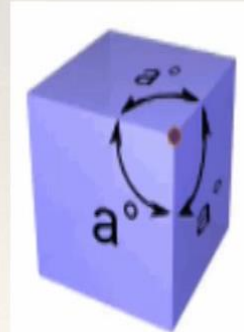
Platonic solids

- ❖ Identify platonic solids and define a solid geometrically
- ❖ Prove that there can only be 5 types of regular solids
- ❖ Identify Lattice parameters for different crystal systems
- ❖ Define a crystal system based on the symmetry level

Platonic solids

- ❖ Tetrahedron
- ❖ Hexahedron
- ❖ Octahedron
- ❖ Dodecahedron
- ❖ Icosahedron

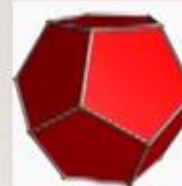
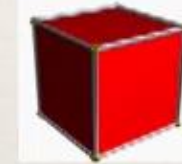
- ❖ A regular solid is formed when more than two regular polygons meet at a corner and the total internal angle at the corner is less than 360°



Regular Convex Polyhedra

Platonic solids

- ❖ Tetrahedron (4 faces)
- ❖ Hexahedron (6 faces)
- ❖ Octahedron (8 faces)
- ❖ Dodecahedron (12 faces)
- ❖ Icosahedron (20 faces)



How many types of regular solids are possible

Geometry	Number	Total angle	Solid
Triangle (60°)	3	180	Tetrahedron
	4	240	octahedron
	5	300	Icosahedron
	6	360	X
Square (90°)	3	270	Hexahedron (cube)
	4	360	X
Pentagon (108°)	3	324	Dodecahedron

- ❖ Tetrahedron (4 faces)
- ❖ Hexahedron (6 faces)
- ❖ Octahedron (8 faces)
- ❖ Dodecahedron (12 faces)
- ❖ Icosahedron (20 faces)



- ❖ Hexagon's internal angle is 120^0 and hence 3 times 120 is 360^0 . Hence, it can't be a solid, but a tiling.



gettyimages
Pongnathee Kuaythong / EyeEm

Crystalline solids

- ❖ Solid materials : regularity of the atomic/ionic arrangement
- ❖ Crystalline material is one in which the atoms (**lattice points**) are arranged in a repeating or periodic array over large atomic distances
- ❖ Long-Range order exists
- ❖ Each atom is bonded to its nearest neighbour
- ❖ Some of the properties depend on the crystal structure
- ❖ All metals, many ceramics and some polymers form crystalline structure under normal conditions

- ❖ Important Physical Properties depend on crystal structure
 - ❖ Conductivity
 - ❖ Magnetic properties
 - ❖ Stiffness
 - ❖ Strength
- ❖ Properties also depend on crystal orientation

Crystalline structure

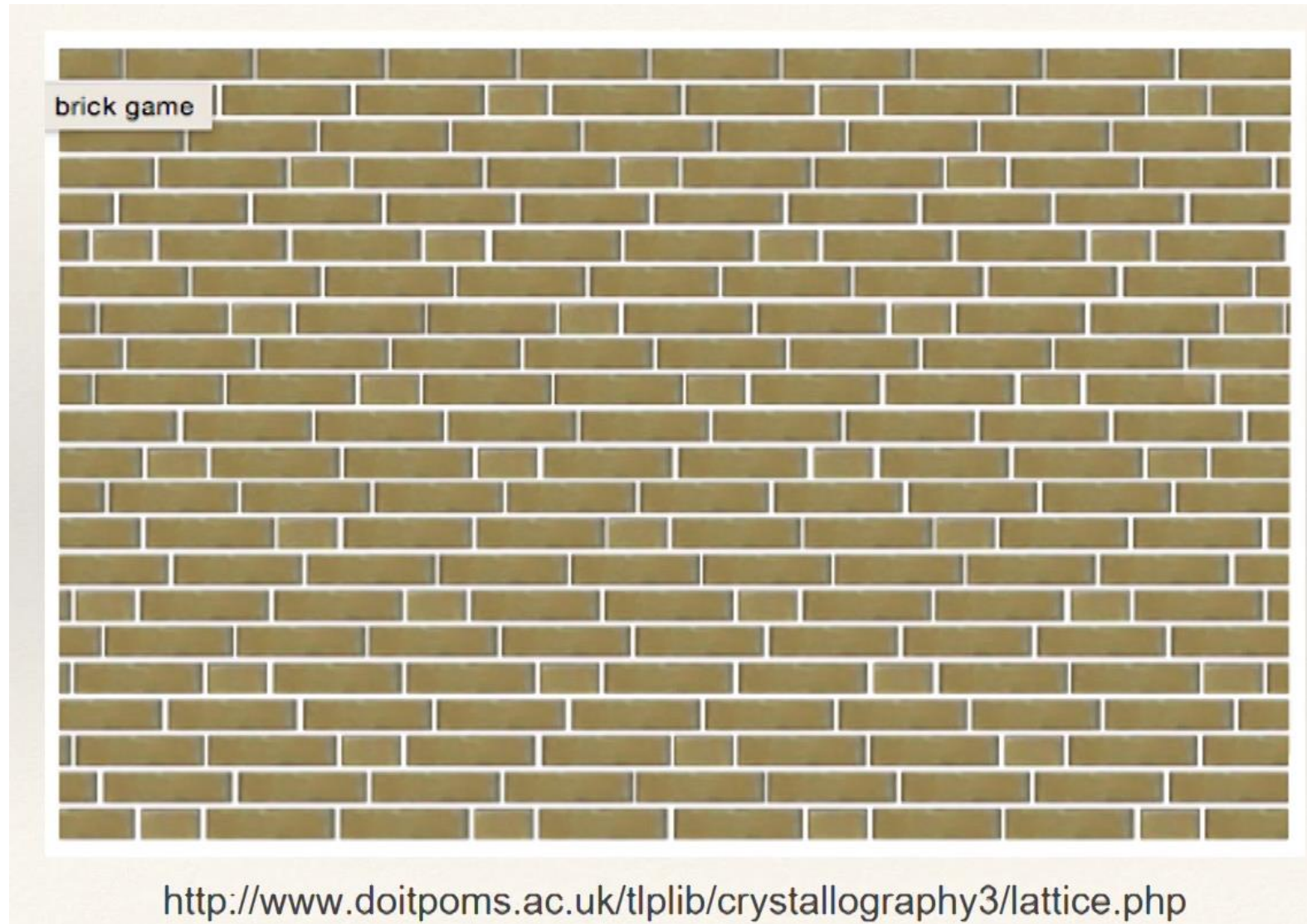
Characterized by repeating structures in 3 Dimensions

- Lattice
- Lattice points
- Unit cell

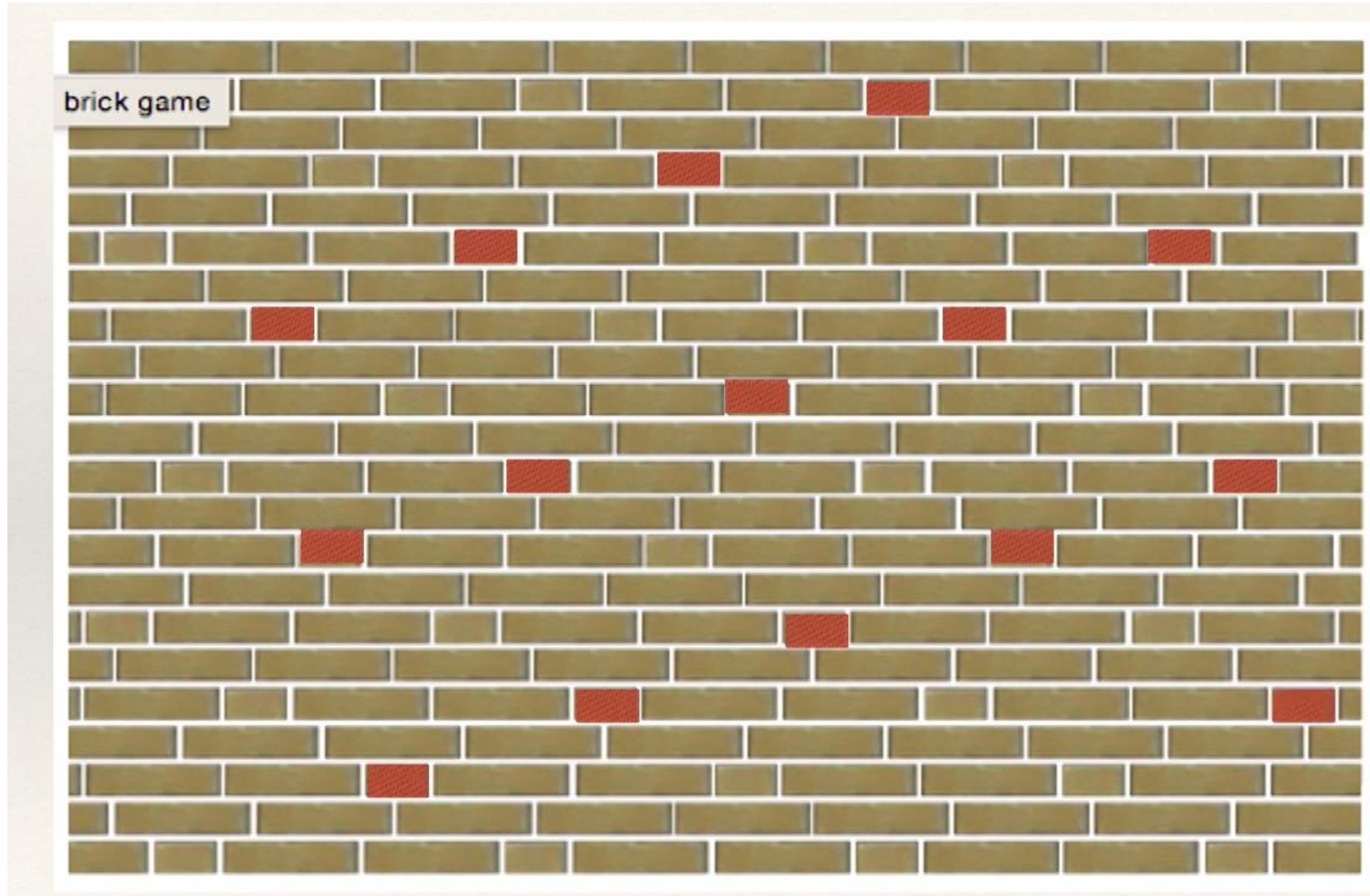
How to identify a lattice ?

- ❖ Choose an arbitrary reference point
- ❖ Mark the points identical to the reference point in the pattern
- ❖ The set of identical points is the **lattice**

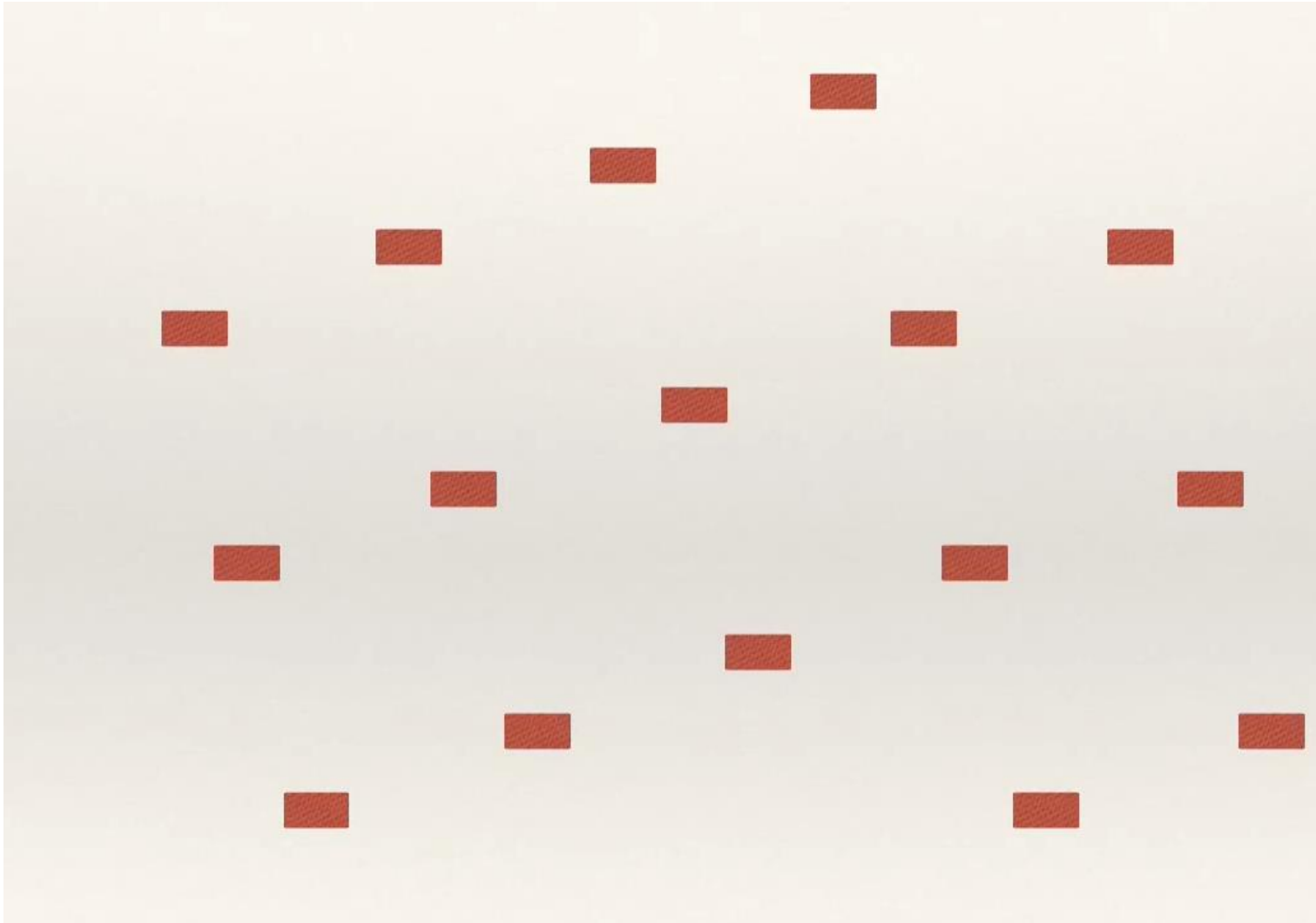
Identify the lattice



One possible 2D lattice (see the red bricks) of the brick pattern example. Other 2D lattices are possible for the same brick pattern.

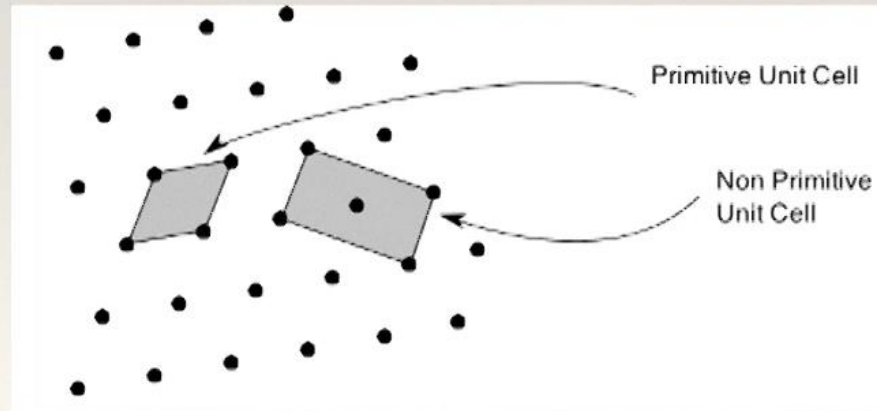


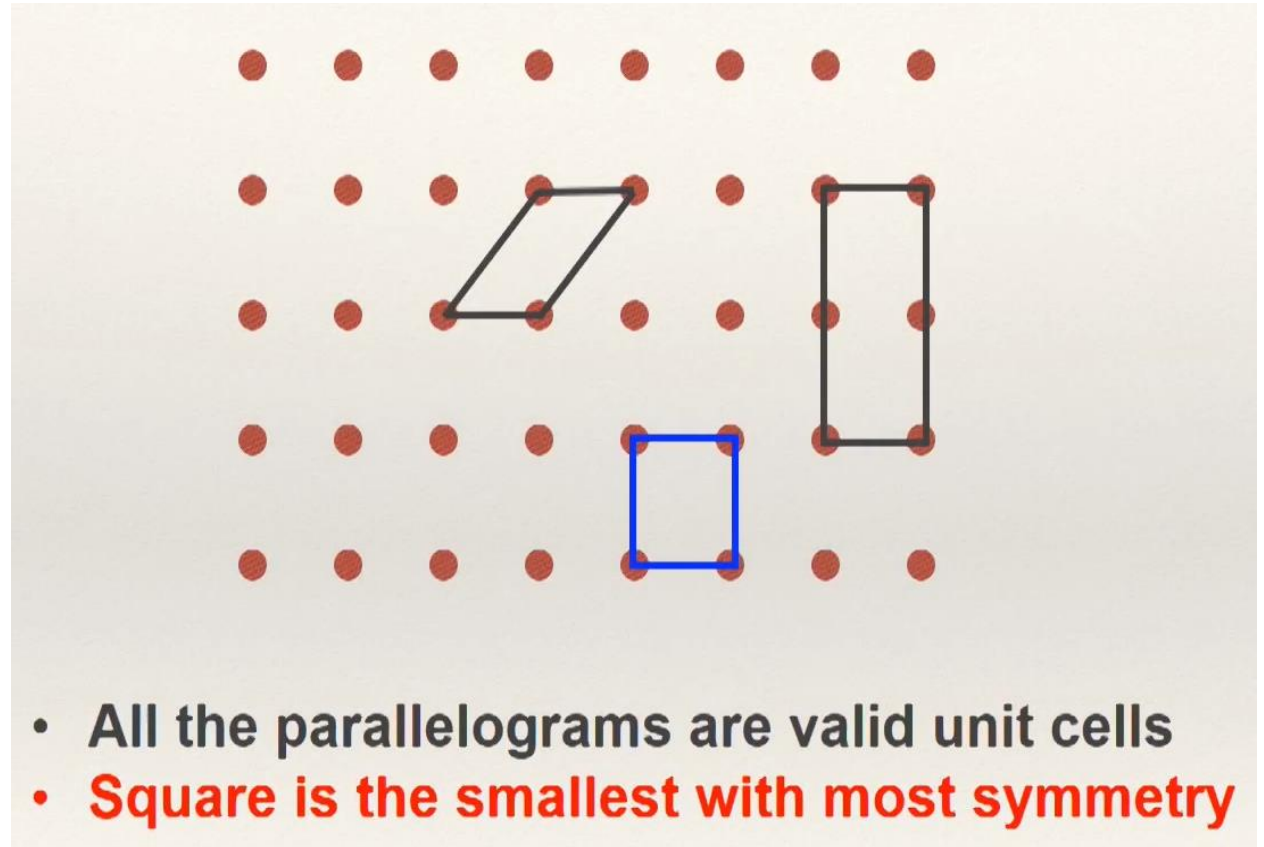
Lattice:- A space lattice is defined as an infinite array of points in three dimensions in which every point has surroundings *identical* to that of every other points in the array.

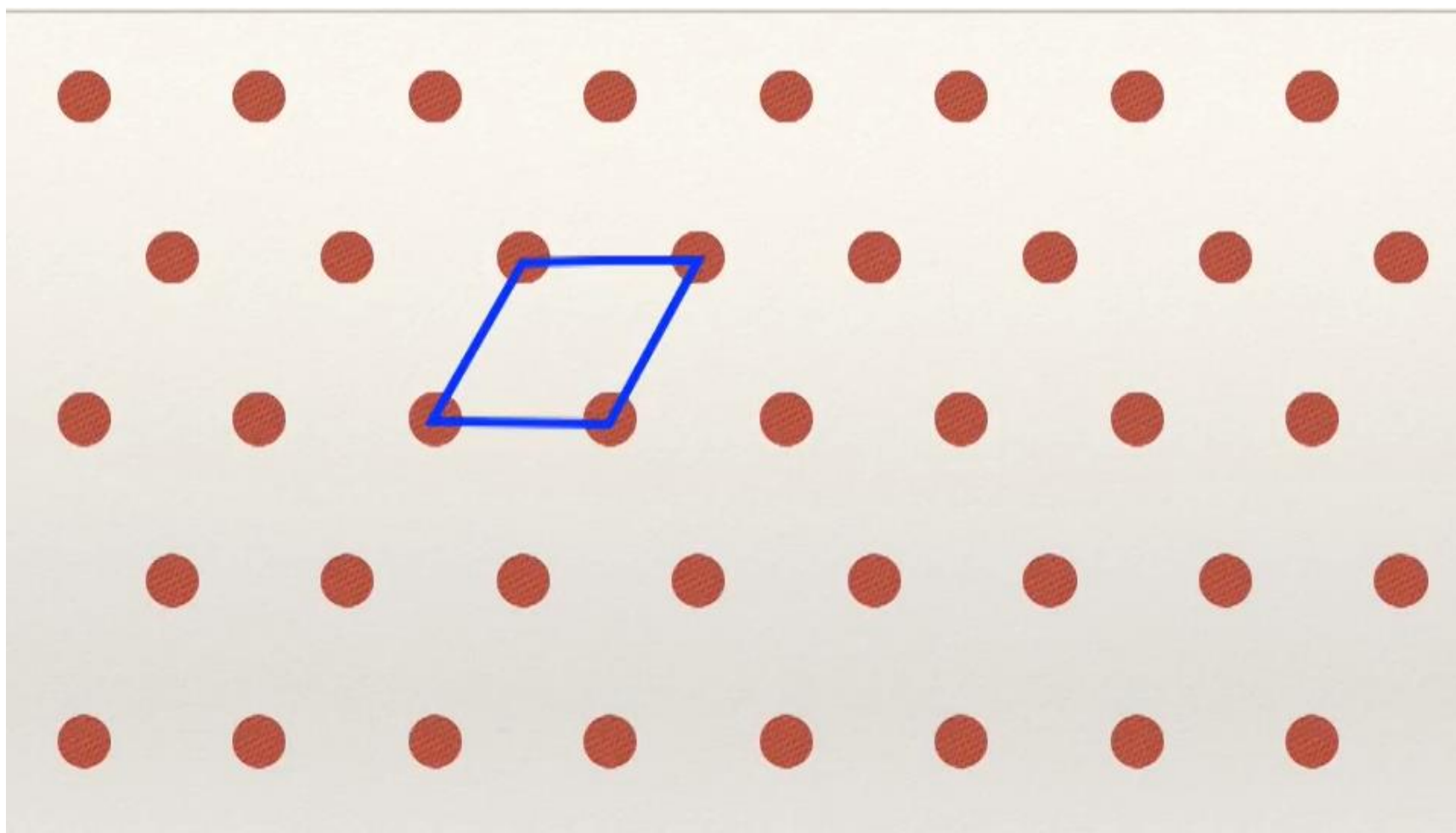


Unit Cell

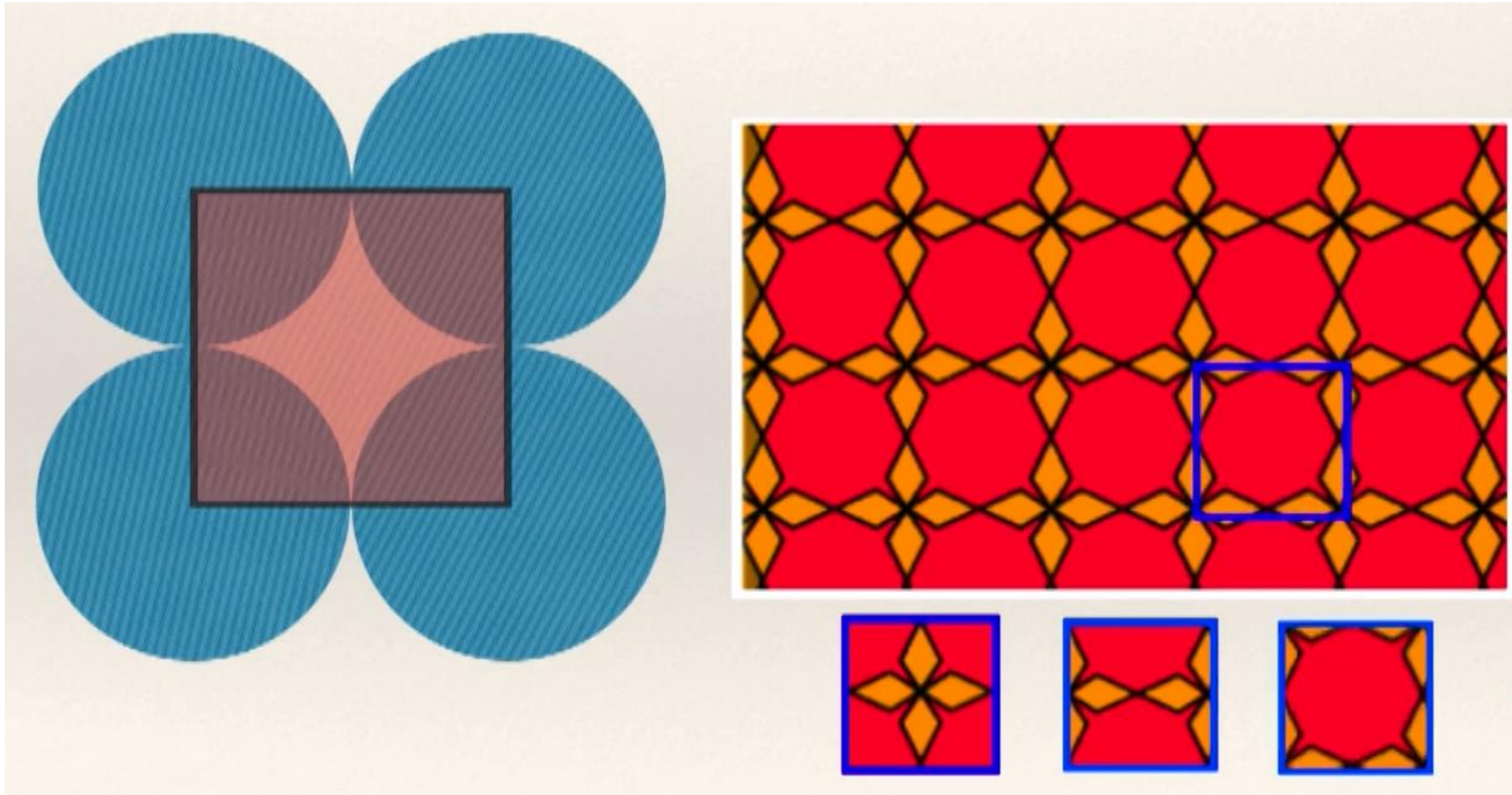
- ❖ Repeated elements (**unit cells**) in three dimensions
- ❖ Unit cells are the building blocks of crystal structure
- ❖ Primitive and non-primitive unit cells
 - ❖ Primitive: only one lattice point per unit cell
 - ❖ non-primitive: more than one lattice point per unit cell







A space lattice can be defined by referring to a unit cell. The unit cell is the smallest unit which when repeated in space indefinitely, will generate the space lattice.



Crystal vs Lattice

Lattice

A translationally periodic arrangement of **points** in 3D

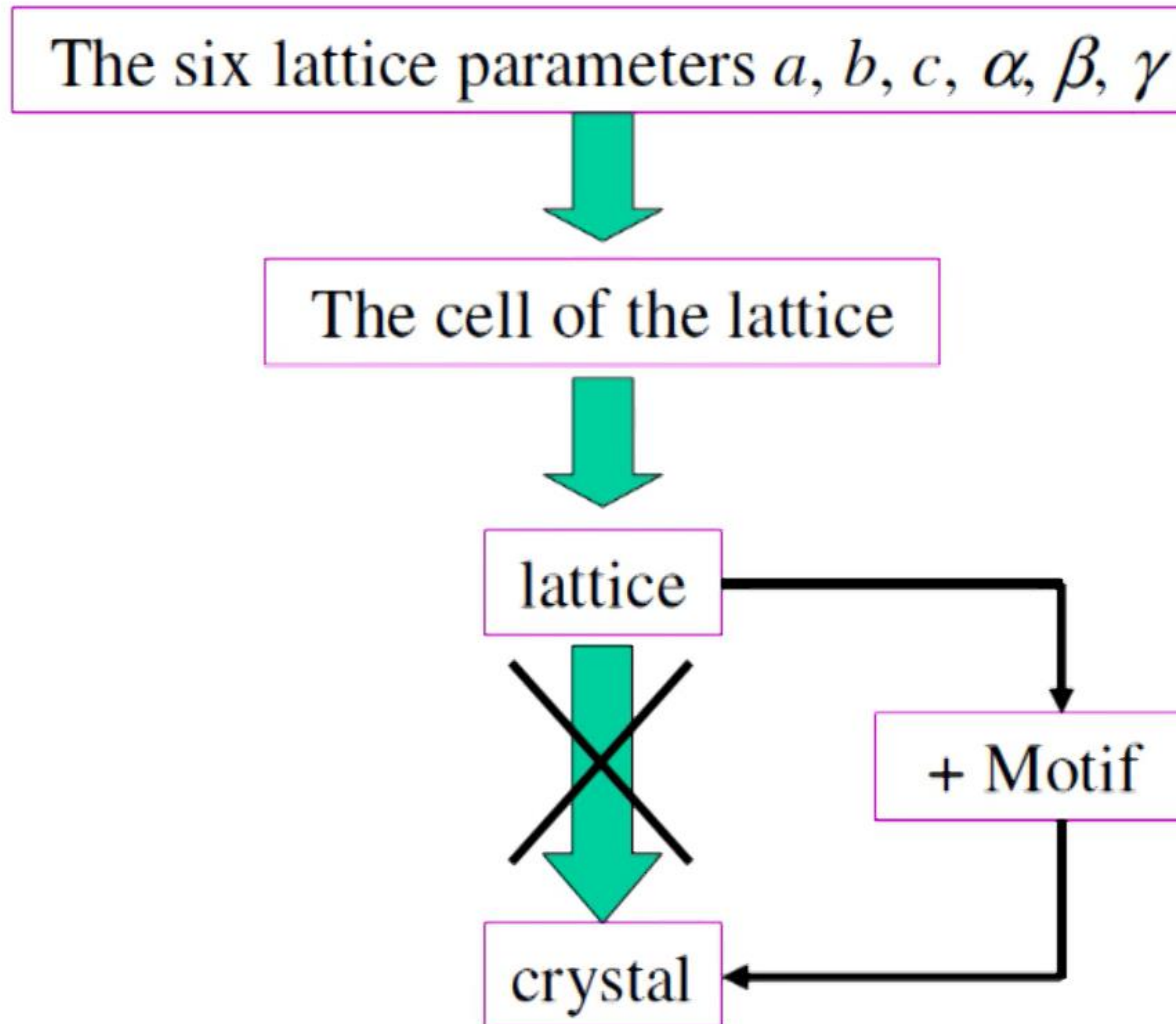
Crystal

A translationally periodic arrangement of **atoms** in 3D

Crystal = Lattice + basis/motif

- Lattice: the periodicity or order of the crystal
- Basis/Motif: atom/group of atoms associated with each lattice point
- Lattice: How to repeat
- Basis: What to repeat

Crystal vs Lattice: A three dimensional space lattice is generated by repeated translation of three noncoplanar vectors $a, b,$ and c .



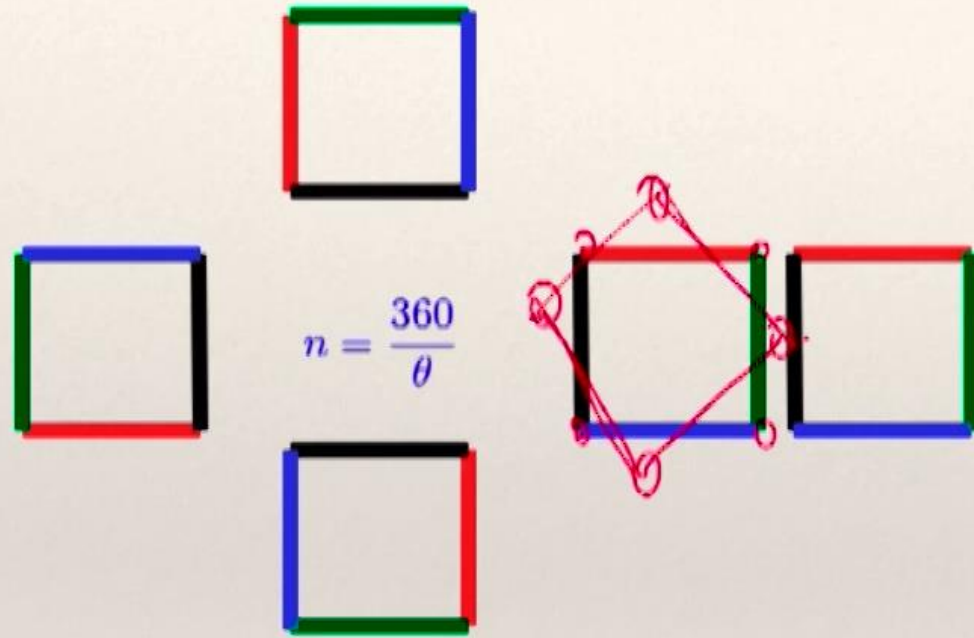
Unit cell

- ❖ In 2D: any parallelogram whose vertices are lattice points
- ❖ In 3D: any parallelepiped whose vertices are lattice points
- ❖ Infinite number (of units cells) possible
- ❖ Smallest unit cell is chosen as a reference

Symmetry of a crystal structure: Crystals have inherent symmetry (rotational, translational and reflection symmetry)

- ❖ Two types of symmetry
 - ❖ Rotational Symmetry
 - ❖ Translational Symmetry
- ❖ A material which has the both types of symmetry
 - ❖ Crystal
- ❖ Materials that do not possess both the symmetries
 - ❖ Amorphous
- ❖ What if it has rotational symmetry and no translational symmetry?
 - ❖ Quasicrystal (special 5-fold symmetry)

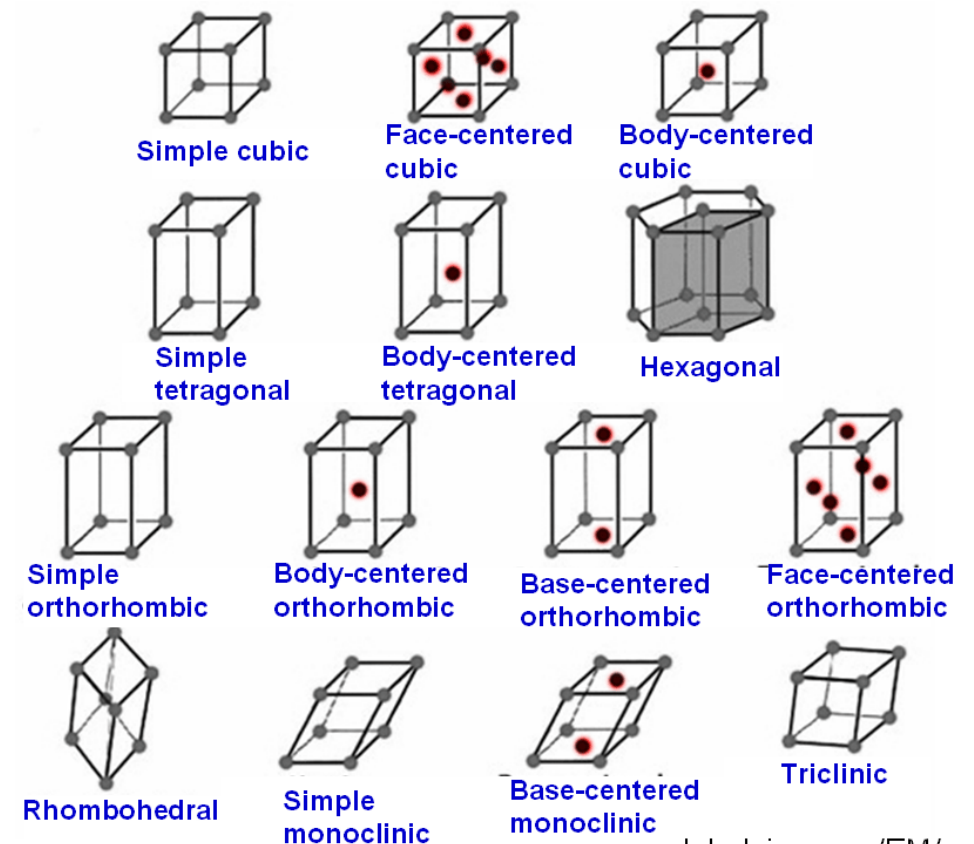
A cube has 3 four fold symmetry



- If we rotate by 45° then we generate new points
- For 90° , we don't generate new points, hence 4 fold symmetry.

Bravais lattices: A three dimensional space lattice is generated by repeated translation of three noncoplanar vectors $a, b,$ and c . It so turns out that there are only 14 distinguishable ways of arranging points in 3D space, such that each arrangement conforms to the definition of a space lattice. These 14 space lattices are known as Bravais lattices, named after their originator). They belong to 7 crystal systems.

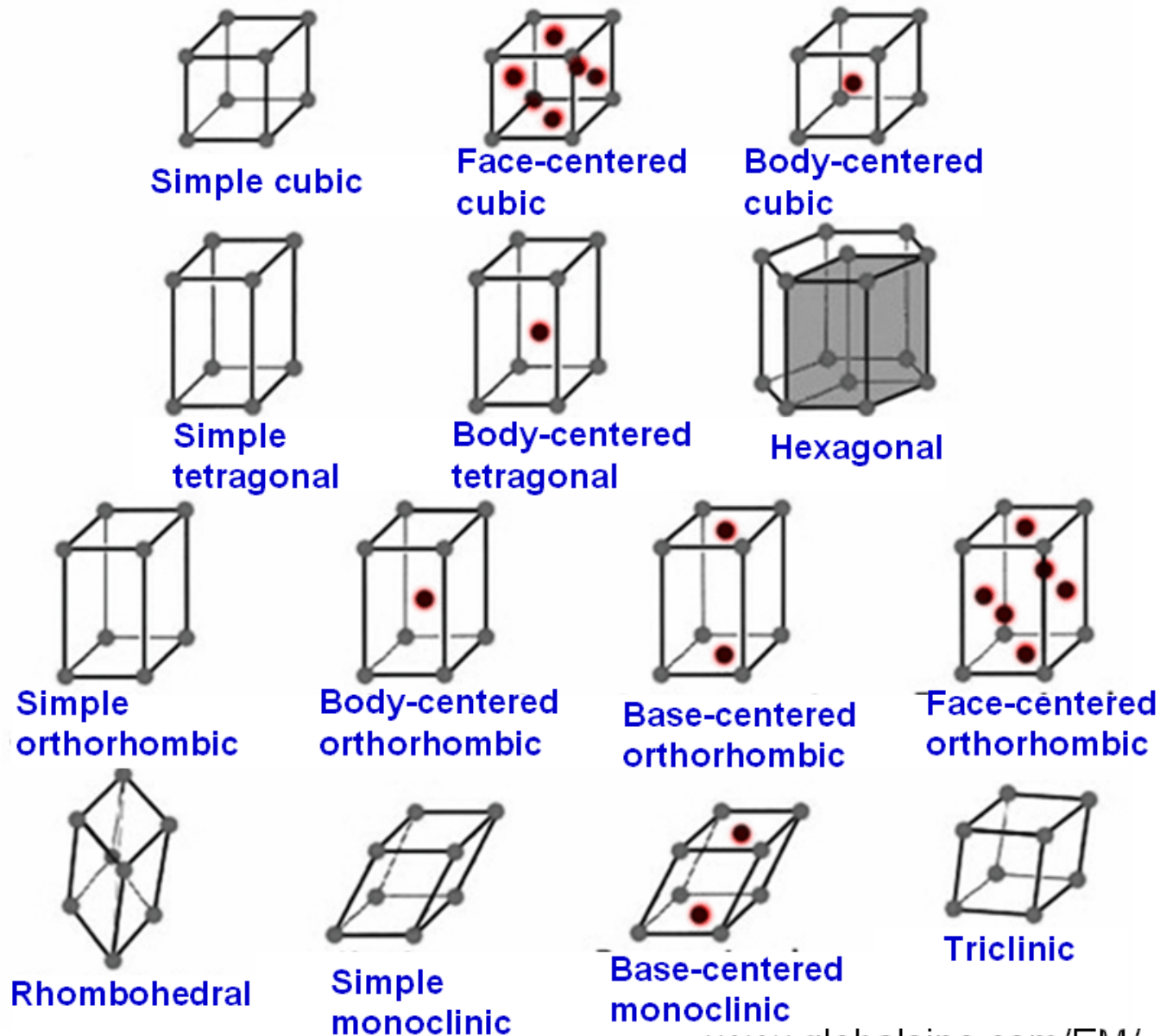
- ❖ Bravais in 1850 has shown that crystals could be divided into 14 unit cells (and only 14)
- ❖ The unit cell is the simplest repeating unit in the crystal
- ❖ Opposite faces of a unit cell are parallel
- ❖ The edge of the unit cell connects equivalent points
- ❖ More than one unit cell may be chosen for a particular crystal structure



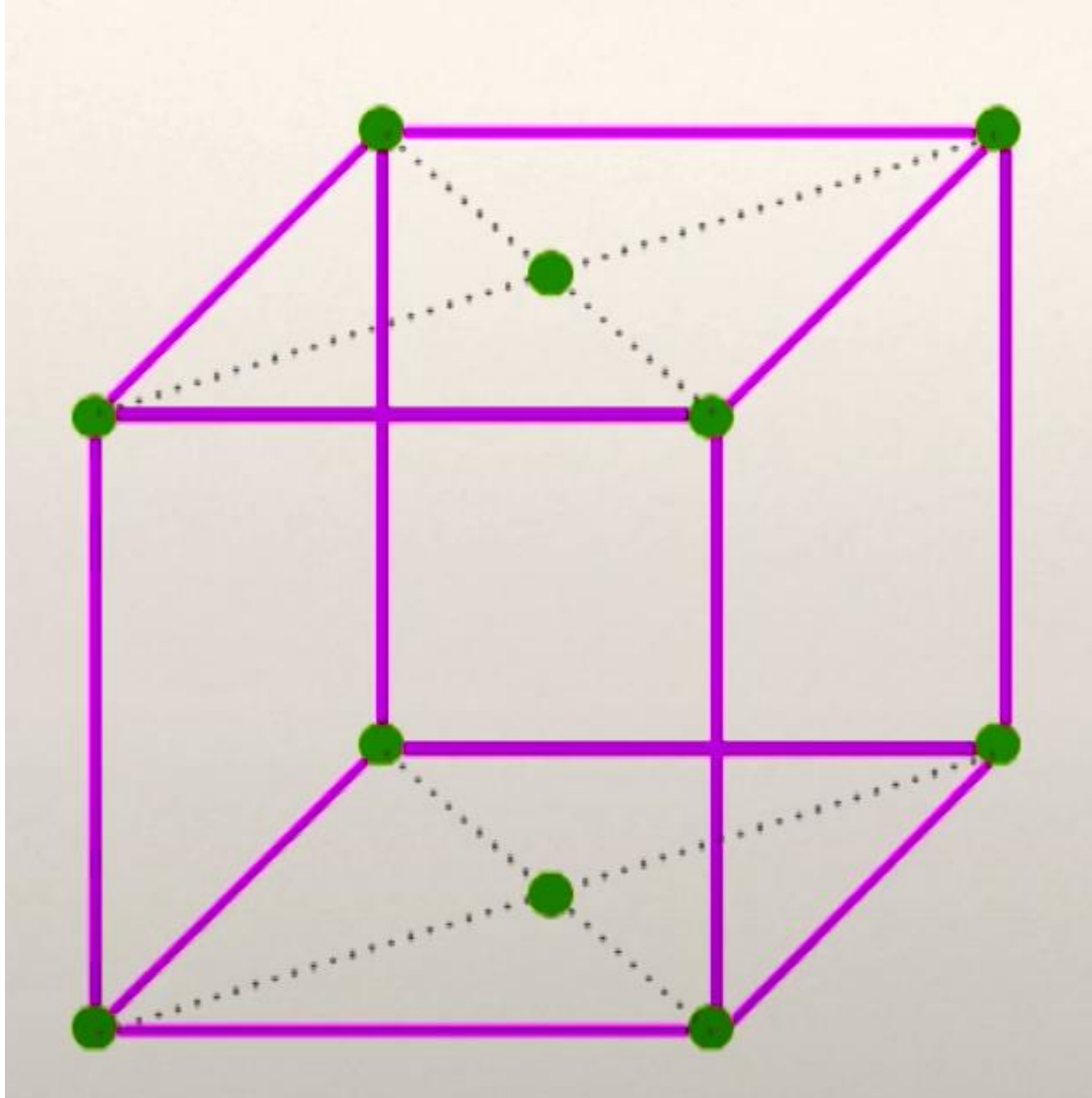
Bravais lattices

- ❖ Cubic ($a=b=c$, all angles= 90°)
- ❖ Tetragonal ($a=b \neq c$, all angles= 90°)
- ❖ Orthorhombic ($a \neq b \neq c$, all angles= 90°)
- ❖ Rhombohedral or Trigonal ($a=b=c$, three equal angles $\neq 90^\circ$)
- ❖ Hexagonal ($a=b \neq c$, two angles= 90° , third angle= 120°)
- ❖ Monoclinic ($a \neq b \neq c$, two angles= 90° , third angle=any)
- ❖ Triclinic ($a \neq b \neq c$, no angle= 90° and none of them are equal)

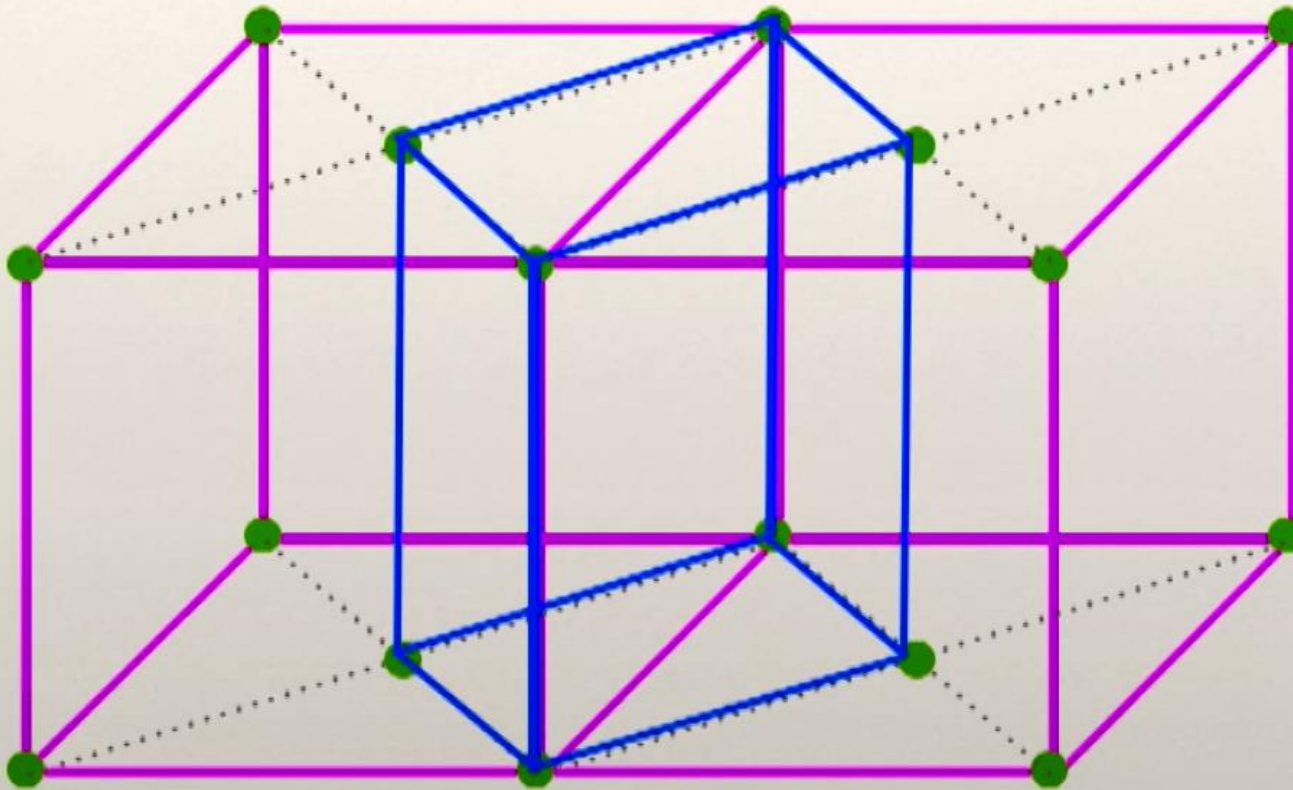
Only 14 ways of filling 3D space with long range order



Why no end centered cubic in Bravais lattice ?



Why no end centered cubic in Bravais lattice ?



Cubic **C** is equivalent to Tetragonal **P**

Symmetry of 7 crystal systems

Crystal system	Required Symmetry Level
Cubic	Three 4-fold axis
Tetragonal	One 4-fold axis
Orthorhombic	Three 2-fold axis
Hexagonal	One 6-fold axis
Rhombohedral (Trigonal)	One 3-fold axis
Monoclinic	One 2-fold axis
Triclinic	None